확률·랜덤프로세스 기말대비 6/12 (D6) · 2차 — 인터리브 복습 L05–L07

① 오답노트.md 어제 오답부터 재풀이 ② 아래 신규 문제는 '상한'(시간 부족 시 신규만 이월) ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

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문항 목차 (총 12문항)

Problem 13

Let X be a random variable with the following CDF

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x < \frac{1}{4} \\ x + \frac{1}{2} & \text{for } \frac{1}{4} \leq x < \frac{1}{2} \\ 1 & \text{for } x \geq \frac{1}{2} \end{cases}$$

a. Plot $F_X(x)$ and explain why X is a mixed random variable.
b. Find $P(X \leq \frac{1}{3})$.
c. Find $P(X \geq \frac{1}{4})$.
d. Write CDF of X in the form of

$$F_X(x) = C(x) + D(x),$$

where $C(x)$ is a continuous function and $D(x)$ is in the form of a staircase function, i.e.,

$$D(x) = \sum_k a_k u(x - x_k).$$

e. Find $c(x) = \frac{d}{dx} C(x)$.
f. Find EX using $EX = \int_{-\infty}^{\infty} x c(x) dx + \sum_k x_k a_k$

— 풀이 —

Problem 16

A company makes a certain device. We are interested in the lifetime of the device. It is estimated that around 2% of the devices are defective from the start so they have a lifetime of 0 years. If a device is not defective, then the lifetime of the device is exponentially distributed with a parameter $\lambda = 2$ years. Let X be the lifetime of a randomly chosen device.

a. Find the generalized PDF of X.
b. Find $P(X \geq 1)$.
c. Find $P(X > 2|X \geq 1)$.
d. Find EX and $Var(X)$.

— 풀이 —

4.5.18♦ In this problem, we outline the proof of Theorem 4.11.

(a) Let X_n denote an Erlang (n, λ) random variable. Use the definition of the Erlang PDF to show that for any $x \geq 0$,

$$F_{X_n}(x) = \int_0^x \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}\, dt.$$

(b) Apply integration by parts (see Appendix B, Math Fact B.10) to this integral to show that for $x \geq 0$,

$$F_{X_n}(x) = F_{X_{n-1}}(x) - \frac{(\lambda x)^{n-1} e^{-\lambda x}}{(n-1)!}.$$

(c) Use the fact that $F_{X_1}(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ to verify the claim of Theorem 4.11.

— 풀이 —

Problem 17
Let X and Y be two jointly continuous random variables with joint PDF

$$f_{XY}(x,y) = \begin{cases} e^{-xy} & 1 \leq x \leq e, y > 0 \\ \\ 0 & \text{otherwise} \end{cases}$$

a. Find the marginal PDFs, $f_X(x)$ and $f_Y(y)$.
b. Write an integral to compute $P(0 \leq Y \leq 1, 1 \leq X \leq \sqrt{e})$.

— 풀이 —

5.4.1 Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} c & x \geq 0, y \geq 0, x+y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c?

(b) What is $\mathrm{P}[X \leq Y]$?

(c) What is $\mathrm{P}[X + Y \leq 1/2]$?

— 풀이 —

5.5.1 Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/2 & -1 \le x \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch the region of nonzero probability and answer the following questions.

(a) What is $\mathrm{P}[X > 0]$?

(b) What is $f_X(x)$?

(c) What is $\mathrm{E}[X]$?

— 풀이 —

5.6.5● X is the continuous uniform $(0, 2)$ random variable. Y has the continuous uniform $(0, 5)$ PDF, independent of X. What is the joint PDF $f_{X,Y}(x, y)$?

— 풀이 —

5.8.2 ● For the random variables X and Y in Problem 5.2.1, find

(a) The expected value of $W = Y/X$,

(b) The correlation, $r_{X,Y} = \mathrm{E}[XY]$,

(c) The covariance, $\mathrm{Cov}[X, Y]$,

(d) The correlation coefficient, $\rho_{X,Y}$,

(e) The variance of $X + Y$, $\mathrm{Var}[X + Y]$.

(Refer to the results of Problem 5.3.1 to answer some of these questions.)

— 풀이 —

6.2.1 ● The voltage X across a 1 Ω resistor is a uniform random variable with parameters 0 and 1. The instantaneous power is $Y = X^2$. Find the CDF $F_Y(y)$ and the PDF $f_Y(y)$ of Y.

— 풀이 —

1.6.1 Is it possible for A and B to be independent events yet satisfy $A = B$?

— 풀이 —

3.7.1 Starting on day $n = 1$, you buy one lottery ticket each day. Each ticket costs 1 dollar and is independently a winner that can be cashed for 5 dollars with probability 0.1; otherwise the ticket is worthless Let X_n equal your net profit after n days. What is $\mathrm{E}[X_n]$?

— 풀이 —

5.2.1 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} cxy & x = 1,2,4; \quad y = 1,3, \\ 0 & \text{otherwise.} \end{cases}$$

(a) What is the value of the constant c?

(b) What is $\mathrm{P}[Y < X]$?

(c) What is $\mathrm{P}[Y > X]$?

(d) What is $\mathrm{P}[Y = X]$?

(e) What is $\mathrm{P}[Y = 3]$?

— 풀이 —